

Project Planning and Business Understanding

Risk Analytics and Claims Intelligence at *AegisLife Insurance Pvt. Ltd.*

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1. Business Context

AegisLife Insurance Pvt. Ltd. is a mid-sized insurance company facing rapid growth in policies, customers, and claims. The management team needs a unified analytics foundation to monitor profitability, claims risk, fraud exposure, operational efficiency, customer risk, and agent productivity.

The major business challenges are:

- ? High claim payouts and loss ratios threatening profitability across products and regions.
- ? Possible claim anomalies and fraud-related patterns requiring stronger monitoring.
- ? Limited unified dashboarding for decision makers across executive, regional, agent, and risk teams.
- ? Need for data-backed recommendations to support pricing, underwriting, reserve planning, and compliance.

2. Analytical Objectives

- ? Build an integrated analytics workflow using SQL, Python, Excel validation, statistics, and Power BI.
- ? Validate data quality and referential integrity across customer, policy, claim, agent, and feedback tables.
- ? Measure core insurance KPIs including premium, claims, loss ratio, approval rate, fraud ratio, policy lapse, and agent productivity.
- ? Identify high-risk segments by region, product type, customer profile, agent performance, and claim status.
- ? Use statistical testing to validate business assumptions rather than relying only on visual patterns.
- ? Develop interactive dashboards and convert insights into actionable business recommendations.

3. KPI Framework

KPI	Formula / Logic	Business Use
Total Premium Collected	Sum of premium earned over policy duration	Measures revenue generated from policy portfolio.
Total Claims Paid	Sum of claim amount approved	Measures financial outflow and claim exposure.
Loss Ratio %	Total Claims Paid / Total Premium Collected x 100	Assesses underwriting profitability.
Claim Approval Rate %	Approved Claims / Total Claims x 100	Tracks claim settlement outcome and process efficiency.
Fraud Ratio %	Fraud-Flagged Claims / Total Claims x 100	Highlights fraud monitoring requirement.
Claim Frequency	Number of Claims / Number of Active Policies or Customers	Measures claim occurrence intensity.
Average Claim Amount	Total Claim Amount / Number of Claims	Measures claim severity and reserve need.
Agent Productivity	Policies Sold or Premium Sold per Agent	Evaluates sales and operating contribution.
Policy Lapse Rate %	Lapsed Policies / Total Policies x 100	Measures retention and policy persistency risk.
Customer Risk Score	Risk score and derived risk segment	Supports underwriting and customer monitoring.

4. Business Questions

- ? Which product types generate the highest claims and loss ratio?
- ? Which regions show profitability concern, high claim payouts, or high fraud exposure?
- ? Are claim outcomes associated with claim type or customer risk profile?
- ? Do smokers, age groups, regions, or policy types have statistically different claim amounts?
- ? Which agents have high productivity and which agents show elevated fraud association?
- ? What is the short-term expected claim exposure for reserve planning?

5. Hypothesis Framework

Hypothesis Area	Null Hypothesis	Recommended Test	Decision Rule
Smoker vs Non-Smoker Claims	No significant difference in average claim amount between smokers and non-smokers.	Independent t-test	Reject H0 if p-value < 0.05.
Regional Claim Amount	Average claim amount is equal across regions.	One-way ANOVA	Reject H0 if p-value < 0.05.
Policy Type Claim Amount	Average claim amount is equal across product/policy types.	One-way ANOVA	Reject H0 if p-value < 0.05.
Age Group Claim Amount	Average claim amount is equal across age groups.	One-way ANOVA	Reject H0 if p-value < 0.05.
Claim Status vs Claim Type	Claim status and claim type are independent.	Chi-square test	Reject H0 if p-value < 0.05.
Risk Score vs Claim Amount	No significant linear relationship exists.	Pearson correlation	Reject H0 if p-value < 0.05.

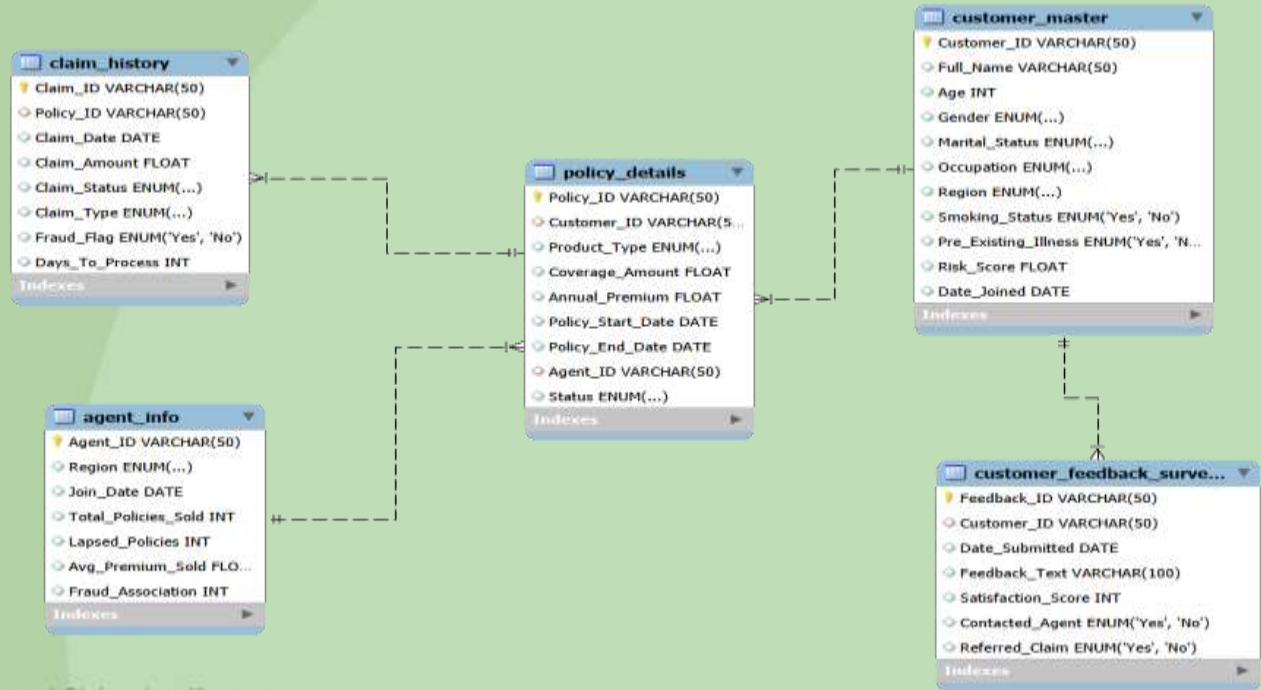
6. Analytics Workflow and Tools

Step	Tool	Output
Business understanding and KPI planning	PowerPoint	Clear objective, KPI framework, hypotheses.
Data validation	Excel	Data Quality Log, FK checks, missing/invalid/outlier flags.
Database schema and ingestion	MySQL	DDL scripts, PK/FK constraints, row-count validation.
Data extraction and wrangling	Python, pandas, SQLAlchemy	Clean analytical dataframes and master analytics table.
EDA and descriptive analytics	Python, seaborn, matplotlib	Trends, distributions, grouped statistics, outlier analysis.
Statistical validation	scipy.stats	Hypothesis tests, p-values, confidence intervals, interpretation.
Dashboarding	Power BI	Executive, claims/fraud, customer risk, regional, agent, forecasting views.
Recommendations	Report	Actionable insights with business impact.

7. Data Scope and Relationships

The analysis uses five linked tables. The policy table links customers and agents, claims link to policies, and feedback links to customers where customer identity is available.

Table	Primary Key	Important Relationships	Analytical Use
Customer_Master	Customer_ID	Referenced by Policy_Details and Customer_Feedback_Surveys	Demographics, region, smoking status, pre-existing illness, risk score.
Policy_Details	Policy_ID	Customer_ID to customers; Agent_ID to agents	Premium, coverage, status, product type, policy duration
Claim_History	Claim_ID	Policy_ID to policies	Claim amount, claim status, fraud flag, processing days.
Agent_Info	Agent_ID	Referenced by Policy_Details	Agent productivity, lapsed policies, fraud association
Customer_Feedback_Surveys	Feedback_ID	Customer_ID to customers when Available	Satisfaction, claim-related feedback, service quality.



8. Expected Deliverables

- ? Business understanding and KPI framework document.
- ? Excel Data Quality Log with validation checks and cleaning actions.
- ? SQL DDL, ingestion validation, row-count checks, and KPI queries.
- ? Python notebook covering extraction, cleaning, feature engineering, EDA, statistics, forecasting, and exports.
- ? Power BI dashboard with executive and role-specific monitoring pages.
- ? Final executive summary report with top insights and strategic recommendations.

9. Success Criteria

- ? Data integrity is validated before analysis and documented transparently.
- ? All KPI definitions are consistent across SQL, Python, and Power BI.
- ? Statistical results include p-values, conclusions, business interpretation, and recommended actions.
- ? Dashboard pages are interactive, clear for non-technical users, and support decision-making.
- ? Recommendations are supported by measurable evidence from the analysis.